MODULE 04 · SECTION 1 - DESIGNING DATA PROCESSING SYSTEMS

Personal Data and Sensitive Data Protection

Find personal data, then hide it in a way that keeps the data useful: tokens, hashes, masks and generalisation.

SECTION 1 of 5 EXAM WEIGHT 22% MODULE 4 of 45

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Hiding personal data is easy. Hiding it **while keeping the data useful** is what the exam tests.

Find it, hide it, check it

01

Discover

Continuous profiles show which tables and buckets hold sensitive data.

02

Inspect

A deep scan finds every occurrence in one table, file or stream.

03

De-identify

Redact, mask, tokenise or generalise, using a reusable template.

04

Measure risk

Check that what is left cannot single people out.

CURRENCY CHECK

Sensitive Data Protection is the new name for Cloud DLP. The API is still dlp.googleapis.com, and the guide says Cloud DLP.

infoTypes: what Sensitive Data Protection looks for

- **Built-in infoTypes** cover common data: EMAIL_ADDRESS, PHONE_NUMBER, IBAN_CODE and many national IDs.
- **Custom infoTypes** add your own patterns: a regular expression, or a dictionary of words.
- Each finding has a **likelihood**, from very unlikely to very likely, so you can set a threshold.
- Larkspur adds a custom infoType for its **meter IDs**, which no built-in type knows.

> WORTH KNOWING
>
> An inspect template saves the infoTypes and threshold, so every job and pipeline uses the same rules.

Discovery or an inspection job?

DISCOVERY

Where is it, across everything?

- Organisation, folder or project scope
- Continuous: new tables are profiled
- Sensitivity and data risk per table
- Sends profiles to Security Command Center

INSPECTION JOB

Exactly where, in this one?

- One table, bucket or dataset
- Finds every occurrence, row by row
- Can save findings to BigQuery
- Run on demand or on a schedule

THE TIEBREAKER

'Across the organisation, as data is added' is discovery. 'Every occurrence in this dataset' is an inspection job.

The de-identification toolbox

METHOD	WHAT YOU GET	REVERSIBLE?	JOINS STILL WORK?
Redaction	The value is removed	No	No
Replacement	A fixed word, or the infoType name	No	No
Character masking	****1234	No	No
Crypto hash	A one-way token, using a secret key	No	Yes
Deterministic encryption	A token that can be decrypted	Yes	Yes
Format-preserving	A token shaped like the original	Yes	Yes
Bucketing, date shifting	Ranges, or shifted dates	No	Partly

THE TIEBREAKER

Two questions pick the method: must anyone ever get the real value back, and must the column still join?

Same input, same key, same token. That is what **keeps your joins alive**.

Pseudonymised or anonymised?

PSEUDONYMISED

Reversible with a key

- Deterministic or format-preserving tokens
- Records still link to one person
- Still personal data under GDPR
- Protect the key like the data

ANONYMISED

No way back

- Redaction, generalisation, aggregation
- Nobody can be singled out
- No longer personal data, if truly so
- Harder to achieve than it looks

THE TRAP

A keyed hash still links one person's records, so treat hashed data as pseudonymised, not anonymous.

Deterministic or format-preserving encryption?

DETERMINISTIC ENCRYPTION

The default choice

- Google's recommended method
- No limits on the input
- Much faster
- Output does not look like the input

FORMAT-PRESERVING ENCRYPTION

Only when shape matters

- Keeps the length and alphabet
- Passes legacy format checks
- Fewer security guarantees
- Needs long enough inputs

THE TIEBREAKER

Choose format-preserving only when something downstream insists on the original format.

Keys for tokens you may reverse

- **KMS-wrapped key.** Your token key is encrypted by a Cloud KMS key. The recommended choice.
- **Transient key.** Generated per request and thrown away. You can never re-identify.
- **Unwrapped key.** The raw key sits in your configuration. Google says do not.
- Re-identification is a separate permission. Most people get tokens; a few get the key.

> ON THE EXAM
>
> 'Must be able to recover the original value later' rules out transient keys and hashing.

Known columns, or text that might contain anything?

RECORD TRANSFORMATIONS

You know the column

- Transform the email column, always
- No inspection needed
- Fast and predictable
- Best for tables and CSV files

INFOTYPE TRANSFORMATIONS

It could be anywhere

- Inspect the text first
- Transform only what is found
- The rest of the text survives
- Best for notes, chats, documents

THE PATTERN

Larkspur's call-centre notes hide bank details in free text. That is an infoType transformation.

A de-identify template for the email column

```
1  {
2    "deidentifyConfig": {
3      "recordTransformations": {
4        "fieldTransformations": [{
5          "fields": [{"name": "email"}],
6          "primitiveTransformation": {
7            "cryptoDeterministicConfig": {
8              "cryptoKey": {"kmsWrapped": {
9                "wrappedKey": "CiQAz7...",
10               "cryptoKeyName": "projects/lk-kms-prod/.../cryptoKeys/sdp"}},
11             "surrogateInfoType": {"name": "EMAIL_TOKEN"}
12         }}}]}}
13 }
```

3 A record transformation: the email column is known, so nothing is inspected.

7-8 Deterministic tokens with a KMS-wrapped key: joins work, and the key team can reverse them.

THE TAKEAWAY

Save it once as a template. Dataflow, jobs and the console reuse it by name.

De-identify on the way in

01

Restricted landing

Raw files in a locked bucket, CMEK, short retention.

02

Dataflow

Calls Sensitive Data Protection with your de-identify template.

03

BigQuery curated

Only tokens and masked values reach the analysts.

04

Original kept

If the business needs it, the raw copy stays locked away.

WORTH KNOWING

Google offers a Dataflow template for exactly this: Cloud Storage CSV to BigQuery, de-identified. It is Beta.

No names, and still identifiable

- **Quasi-identifiers** such as postcode, birth date and gender can single people out when combined.
- **k-anonymity**: every record matches at least k minus one others on those columns.
- **l-diversity**: each such group also holds at least l different sensitive values.
- Fix a low score by **generalising**: a postcode district instead of a full postcode, age bands instead of birth dates.

> ON THE EXAM
>
> Sensitive Data Protection's risk analysis computes these for a BigQuery table before you share it.

Which tool for which privacy job

THE QUESTION WANTS	REACH FOR
Where is personal data, **across the organisation**?	Discovery profiles
Every occurrence in this dataset	An inspection job
Join on a value nobody may see	Deterministic tokens, or a keyed hash
Recover the original later	Deterministic encryption, KMS-wrapped key
Some readers see real values, others masked	BigQuery dynamic data masking, module 29
Erase **one person** everywhere	AEAD keysets, module 03
Share without **re-identification**	Risk analysis, then generalise

THE TIEBREAKER

If two options both hide the data, pick the one that keeps the use the question cares about.

> Larkspur's analysts must join billing customers to app sign-ups on email address, but must never see a real address.
>
> Occasionally the fraud team needs to recover the real email for a flagged account.

How should the email field be de-identified in both datasets?

A Replace email with a keyed crypto hash in both datasets

B Deterministic encryption with the same KMS-wrapped key in both datasets

C Redact the email field before either dataset is loaded

D Mask every character of the email except the domain name

◆ PAUSE AND DECIDE

How should the email field be de-identified in both datasets?

A **Keyed crypto hash**

Joins would work, but a hash can never be reversed. The fraud team could not recover anything.

B **Deterministic encryption, same wrapped key**

Same email, same token, so joins work. The key team can decrypt one flagged token.

C **Redact the field**

Nothing left to join on, and nothing to recover.

D **Mask all but the domain**

Thousands of customers would share one masked value, so the join silently matches the wrong people.

THE SIGNAL

Join plus recover means a reversible, consistent token. Join alone would allow a hash.

> Larkspur now has about 400 projects. The security team must know which BigQuery tables and Cloud Storage buckets hold personal data, including tables created next month.
>
> Results must appear in Security Command Center, with the least ongoing effort.

What should the security team set up?

A. A scheduled inspection job for each existing table, run by Cloud Scheduler

B. A weekly Dataflow job that samples every table and writes a report

C. A request for each data owner to tag sensitive columns in Knowledge Catalog

D. Organisation-level discovery, publishing profiles to Security Command Center

◆ PAUSE AND DECIDE

What should the security team set up?

A **Scheduled inspection jobs per table**

Covers only tables that exist today, and someone must add a job for every new one.

B **A weekly Dataflow sampler**

Custom code to build and maintain, when a managed feature already does this.

C **Ask owners to tag columns**

Depends on people remembering. Untagged data is exactly the risk.

D **Organisation-level discovery to Security Command Center**

Continuous, covers new tables automatically, and publishes where security already looks.

THE SIGNAL

'Across the organisation', 'including new tables' and 'least effort' is discovery.

> Call-centre agents have pasted customers' IBANs into a free-text notes column in Oracle billing. The data reaches BigQuery's raw dataset through Datastream.
>
> Analysts read the notes in the curated dataset. Bank details must be removed, but the rest of each note must stay readable.

What should the pipeline that builds curated do?

A Inspect notes for IBAN_CODE and replace each finding with its infoType name

B Apply a record transformation that redacts the whole notes column

C Put a column-level policy tag on notes so analysts cannot read the column

D Apply format-preserving encryption to the whole notes column

◆ PAUSE AND DECIDE

What should the pipeline that builds curated do?

A Inspect for IBANs and replace each finding

Only the bank details change, to the word IBAN_CODE. The rest of the note survives.

B Redact the whole column

Removes the bank details, and every useful word with them.

C Column-level policy tag

Hides the whole column from analysts. They lose the notes they need.

D Format-preserving encryption on the column

Encrypts the entire note, so nothing is readable. FPE suits short structured values.

THE SIGNAL

'Free text' plus 'keep the rest readable' is an infoType transformation, never a whole-column one.

A university wants Larkspur's half-hourly consumption by full postcode for a heat-pump study. Names, emails and account numbers are already removed.

Some postcodes cover only two or three homes. Larkspur must be confident no household can be singled out.

What should Larkspur do before sharing?

A Share as it is, since nothing left in the data is a direct identifier

B Encrypt the postcode deterministically and give the university the key

C Run a k-anonymity risk analysis, then generalise postcodes until groups pass

D Replace each postcode with a keyed crypto hash before sharing

◆ PAUSE AND DECIDE

What should Larkspur do before sharing?

A Share as it is

A postcode with three homes plus a consumption pattern can point to one household. Quasi-identifiers.

B Encrypt, and share the key

With the key, the university simply decrypts the postcode. Nothing is protected.

C Risk analysis, then generalise

Measures the real risk, then widens postcodes to districts until no group is too small.

D Hash each postcode

A three-home group is still a three-home group, just with a different label. The risk is unchanged.

THE SIGNAL

Removing names is not anonymising. Small groups need generalisation, and risk analysis tells you how far.

Module 04 in five lines

- The exam asks how to hide personal data **and keep it useful**.
- **Discovery** for across the organisation. **Inspection jobs** for every occurrence here.
- **Join and recover**: deterministic encryption, KMS-wrapped key. **Join only**: a keyed hash.
- **Known column**: record transformation. **Free text**: infoType transformation.
- **No names is not anonymous.** Measure risk, then generalise quasi-identifiers.

COMING UP

Next: module 05, residency, perimeters and environments.